

CAD/PAD AMMUNITION DATA CARD MASTER (Rev 16)

13 June 2023

* * * * AMMUNITION DATA CARD * * * *

NSN: 1377009991212 M009 [NSN and DODIC with no dashes.]
 Nomenclature: CARTRIDGE IMPULSE, MK 99 MOD 1 [Include model; if applicable; see Contract Section B.]
 Lot Number: PQP06A001-008 [In accordance with MIL-STD-1168; see Contract Section D.]
 Contractor: PROVEN QUALITY PRODUCTS INC.
 Net Quantity: 2722 [Qty less LAT, investigative (retains), and rejects. Overage units are included. For FAT lots, put "N/A" or "0."]
 Packing: SPI 01-0633165H [List SPI or DWG, if applicable, otherwise a brief description; see Contract Section D]
 Contract/Order Num: N0010406CK999 [Prime contact number with no dashes. See REMARKS for IDIQ contract guidance.]
 Purchase Order Num: [Subcontractor purchase order number from the prime; if applicable.]
 Drawing Number: 951AS111 [Use Government number; if none use contractor P/N; see Contract Section B]
 Drawing Tab: *
 Drawing Rev: B [If none, put "NR" for no revision.]
 Spec Num: MIL-DTL-99999 [If none, then add approved ATP.]
 Spec Rev: C [If none, then add revision of approved ATP.]
 Spec Amend: 1 [If none, put "N/A"]
 Date Started: 01/01/2006 [Format: MM/DD/YYYY. Month must match month letter in lot no. Date production started.]
 Date Completed: 04/20/2006 [Format: MM/DD/YYYY. Date lot was completed.]
 Date Inspected: 05/01/2006 [Format: MM/DD/YYYY. Date lot presented to DCMA QAR for inspection.]
 Line: *
 Zone Weight: *
 Charge Weight: 0.000260 [If end item contains propellant, enter total weight in pounds for a single unit.]
 Index of Powder: *
 Max Pack Depth (US): *
 Max Pack Depth (Metric): *
 Prod Pack Depth: *
 Net Explosive Wt (US): .00106 [Net explosive weight (in pounds) of a single unit; see contract Section D]
 Net Explosive Wt (Metric): .000481 [Net explosive weight in kilograms of a single unit.]
 Test Samp: 47 [Total number of LAT or FAT samples including investigative samples (retains).]
 Test Site: 4 [Test activity code where LAT or FAT is conducted; dropdown menu option.]
 TS Date Shipped: ** 1/11/2018 [Format: MM/DD/YYYY]
 TS Mode Shipped: FEDEX [Method of Shipment]
 UN Proper Shipping Name: CARTRIDGE, POWER DEVICE [See Contract Section D.]
 Hazard Class: 1.4C [See Contract Section D.]
 Gov't QA Activity: 039 [DCMA activity code found in WARP Manual page 58-61]
 Disposition: PA [PA – Pending Acceptance for LAT/FAT, AC – Accepted Lot (from NAVSUP Approval Letter)]
 Date Signed: 06/25/2006 [Format: MM/DD/YYYY. Date Signed by Government Inspector.]

* * * * COMPONENTS * * * *

[List all components, not just explosives/energetics. It should include explosive subassembly components down to the raw material level when not covered by a separate ammunition data card. Energetic raw material and mixes must list the original manufacturer and not a distributor.]

	COMPONENT	DRAWING NO	REV	MANUFACTURER	DATE	LOT NO	QTY MANUFAC TURED
1	O-RING	MS3324-8	C	BARKER	1205	23025	2775
2	ADHESIVE	MMM-A-134	B	4M SEALANTS	0405	36669	AR
3	PRIMER MIX 878	2240790	D	PQP INC	0305	PQP123	2775
4	PROPELLANT	855AS223	D	ACME INC	1205	ACM09	2.5 LB
5	CLOSURE DISC	875AS444	M	ALVIN CORP	0805	33217	350 GM

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6	PIN, FIRING	2514823	K	ILKKA INC	0605	35889	2775
7	SPRING	5558990	A	MARGIOTTA SPRINGS	0705	77899	2775
8	PIN, SHEAR	838AS432	Q	REIMER PINS	0705	99787	2775
9	SET SCREW	2246789	R	HERNANDEZ SCREWS	0505	06778	5550
10	RING, LOCK	777AS777	L	FRODO INC	0605	87877	2775
11	BODY, CARTRIDGE	877AS888	W	KRIKSTAN INC	1105	55676	2775
12	DROID	C-3PO	NR	TATOOINE CO	0605	15664	2775
13	PRIMER	877AS999	AA	NSWC IHD	0904	002158	2775

**** REMARKS ****

[List useful lot information not accounted for in other ADC fields.]

[For IDIQ contacts, list the delivery order in the remarks. See example below.]

[An accurate and understandable accounting of entire lot by serial number is required. Overage/attrition units shall be captured by serial number. Tensile test specimens for linear products shall be identified by serial number.]

	REMARK
1	DELIVERY ORDER N0010417FE001 [For IDIQ contracts only.]
2	(LOT S/N RANGE: 000001 THRU 002775) [Up to eight numeric characters IAW CAD/PAD Marking Instruction.]
3	LAT/FAT SAMPLE S/N: 000002, 000022, 000034, 000145, 000179, 000222, 000345, 000478, 000666, 000788, 000998, 001005, 001035, 001165, 001245, 001356, 001599, 001655, 001788, 002190, 002001, 002069, 002100, 002267, 002345, 002544, 002699, 002701, 002722
4	RETAIN S/N: 000003, 000033, 000200
5	REJECT S/N: 000175, 000435, 001050, 002233
6	OVREAGE/ATTRITION S/N: 002589, 002650 [if applicable]
7	LOT REWORKED FOR SHEAR PINS PER APPROVED REWORK PROCEDURE PQP-RW06

**** WAIVERS AND DEVIATIONS ****

[List all approved request for variances (RFV) that are applicable to the manufactured lot along with a brief description.]

	RFW/RFD	DESCRIPTION
1	RFV FY06-V001	SIX UNITS FAILED PMAX REQUIREMENT OF 3100 PSI WITH WORST CASE BEING 3099 PSI
2	RFV FY06-V002	ALLOW THE USE OF ALTERNATE MATERIAL FOR CARTRIDGE BODY

Additional Ammunition Data Card Helpful Hints/Reference Notes

The above is made up of different lot scenarios to provide example. This is meant to be used as a guide to help remove frustration and provide some consistency for the inputter and the reviewers. This guidance is not meant to cover/address all situations. For more details, go to the Worldwide Ammunition Repository Program (WARP) User's Manual on the WARP website below. If still unsure, seek additional guidance from the acquisition management specialist listed in the contract.

Access: Currently (as of this document's revision date) to access WARP, you must purchase an external certification authority (ECA) through a Government-approved vendor such as IdenTrust (<http://identrust.com/warp/>).

Notice: WARP has been mandated to move to an alternate multi-factor authenticator. As of the date of this document, no formal date has been established. When this change occurs, users will no longer be able to use an ECA. Users will be required to use one of the following methods for access:

- EAMS-A MobileConnect: Download the MobileConnect app to a personal mobile device via Google Play or Apple App store.
- Yubikey: Purchase a Yubikey hardware token that supports FIDO1/WebAuthN (<https://yubico.com/store/compare/>).

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Prior to registering either EAMS-A MobileConnect or Yubikey, each user will require a sponsored EAMS-A account. To request an account, email the user's email address, first name, middle name, and last name to usarmy.ria.jmc.mbx.warp@army.mil. Once an EAMS-A sponsored account is established, proceed with registering via the chosen method:

- a. EAMS-A MobileConnect: <https://mobileconnect.us.army.mil/>
- b. Yubikey: <https://fido2.federatoin.eams.army.mil/yubikey/register>

Once registered, go to the WARP website. Follow the instructions for multi-factor authentication process. Click on "Help" and follow the instructions for obtaining the necessary access.

WARP Website: <https://mhpwarp.redstone.army.mil>

The ammunition data card serves as a "birth certificate" for the lot. It gives all the details of the manufactured lot at fingertips. It is used for malfunction investigation and ammunition reclassification. Incorrect or incomplete information on ADC's come down to a safety issue.

The ammunition data card is not intended to identify specific shipments within a given lot.

* Denotes fields not required for CAD/PAD

** Ammunition data cards that are "PA" do not require a shipped date (Only "AC" ammunition data cards after NAVSUP lot release letter)

Abbreviations: LB = pounds, GR = grains, GM = grams, AR = as required

Difference between the FAT and/or LAT and production lot ADC – The only difference between the FAT and/or LAT and production lot ADC should be the disposition line from "PA" to "AC;" unless there are rejected serial numbers, request for variances (RFVs) and minor corrections that need to be added in the appropriate sections.

Explosive Subassemblies – Cartridges, initiators, etc. that are manufactured by a sub-vendor/subcontractor shall have full traceability. The prime contractor shall be held responsible for traceability of all piece parts associated with the build. Traceability shall be provided as one of the following options:

1. The prime contractor can provide the full breakdown listing of all piece parts that comprises the explosive subassembly. The listing shall include the part/drawing numbers, revisions, manufacturer names, lot/batch numbers, quantities and dates of manufacture. The prime contractor shall enter the listing information into the components section of the end item ADC.
2. The prime contractor can enter the explosive subassembly as a single line on the ADC for the end item. The listing shall include the part/drawing number, revision, manufacturer name, lot/batch number, quantity and date of manufacture of the explosive subassembly. The prime contractor shall retain a copy of the full breakdown listing of the explosive subassembly for their records and be made available upon request by the Government.

USAF & USN CAD/PAD STATEMENT OF WORK - AMMUNITION DATA CARDS

January, 2018

1. Ammunition Data Cards (ADC) shall be prepared in accordance with MIL-STD-1168 (current revision) and shall follow the format required by the application identified as Worldwide Ammunition-data Repository Program (WARP). All of the requirements of MIL-STD-1168 apply but some are tailored in accordance with MIL-STD-962 to meet CAD/PAD program requirements as listed below.
2. The criteria and procedures for the assignment of lot numbers for energetic materials (pyrotechnics, propellants, explosives) including their constituents shall be in accordance with MIL-STD-1168.
3. All energetic materials (pyrotechnics, propellants, explosives) used in assemblies and sub-assemblies/components (i.e., igniter cartridges, delay elements, primer assemblies, explosive cord, etc.) shall be entered in the ADC using the common name, actual/original manufacturer's part number & revision, lot number, and date of manufacture. Traceability shall be ensured by the original manufacturer's documentation such as Certificate of Conformance (COC), Certificate of Analysis (COA), Quality Assurance Record (QAR) Quality Inspection Record (QIR), Quality Assurance Documents (QAD), Batch Record, or other equivalent documentation that would provide positive traceability in the event of a future product recall.
4. The contractor may list their own internal part number for energetic material that they re-designate or re-characterize but must also list the actual manufacturer's part number, lot number, and date of manufacture in parentheses in a format that is searchable in WARP. Any GFM/GFP provided under contract to the vendor should retain government provided part/lot numbers and nomenclature to retain traceability. Ingredients which are used to synthesize or that are blended in a process to make the end item energetic material are not required to be listed. Examples of blended or synthesized energetic materials that shall be listed are HNS, Lead Azide, BKNO₃, ZPP, HES 8028, etc. If the ages of energetic materials are older than the ages specified on the Airmunitions Specification or as specified elsewhere in the contract at the time the CAD/PAD devices become hermetically sealed, the date of energetic material recertification or any associated approved Request for Variance (RFV) shall be notated in the waivers and deviation description section.
5. Hardware components shall be fully traceable and identified by the original manufacturer's lot identification information/documentation or qualified distributor's documentation such as Certificate of Conformance (COC), Certificate of Analysis (COA), Quality Assurance Record (QAR), Batch Record, or other equivalent documentation that would provide positive traceability in the event of a future product recall. The actual date of manufacture is not required for any non-age degradable materials such as metal housings, tubes, sleeves, firing pins, nuts, bolts, washers, etc. The COC, COA, QAR, or Batch Record date(s) may be used for the DOM. Epoxies, sealants, curative, O-rings, and other organic materials shall be fully traceable to OEM with DOM and traceable via the original manufacturer's lot identification documentation such as Certificate of Conformance (COC), Certificate of Analysis (COA), Quality Assurance Record (QAR) Batch Record, or other equivalent documentation that would provide positive traceability in the event of a future product recall. Materials or components that do not affect the reliability or safety of the end item (i.e., solvents, paints, lacquers, labels, dust caps, etc.) do not need to be listed on the ADC.

ATTACHMENT 2 (continued)

6. The ADC data and information entered in WARP for all materials and components ideally shall stand alone for traceability to the actual manufacture's information (i.e. Mfg P/N, Mfg date (as applicable), Mfg lot number etc.). If contractor uses their own P/N, reference date for acceptance or other "trace"/traceability identification used internally for the non-critical piece parts/materials, it shall be noted in the remarks section, i.e. "trace ID, or PO used for the following piece parts: (parts shall identified/listed)", provided the field/entry method is searchable in WARP. If traceability documentation is "questionable" or in conflict and needs clarification, the applicable engineer or equipment specialist shall be contacted to resolve any issues.
7. A "sample" ADC is generated in WARP the first time an ADC is submitted by the contractor for applicable government reviews and approvals. Upon approval of the sample ADC, a final version for each production lot shall be completed in WARP and uploaded for Invoicing, Receipt, Acceptance, and Property Transfer (iRAPT - formerly WAWF) for review and acceptance prior to shipment of product by the applicable government authority. The WARP ADC program will not allow the submission of additional ADCs until such time as the sample ADC has been approved in the system. The sample ADC does not require the use of a "Z" interfix number used to identify it as a sample.
8. If any unique situations exist that may conflict with compliance, a resolution should be proposed by the contractor prior to contract award to determine possible work-around.
9. A hard copy of the applicable ADC(s) shall also be included with each shipment per applicable ADC CDRL.

WARP will reside within the Munitions History Program (MHP). Additional details on these WARP applications are provided below.

MHP-WARP Access Procedures

- Government or Contractor employee with CAC and AKO account:
 - (a) Click on the MHP hyperlink which is <https://mhp.redstone.army.mil/>
 - (b) Enter CAC PIN when prompted
 - (c) Click on WARP (ADC)
 - (d) Click on Help
 - (e) Click on WARP Request Access and follow instructions
- Contractor or Government employee without CAC and AKO account: MHP-WARP uses PKI authentication requiring a DoD approved digital certificate as a security measure to protect the integrity of stored data. There are three vendors that have been approved to issue DoD approved certificates per an External Certification Authority (ECA) program. You are required to use one of the approved vendors listed on the following DISA website: <http://iase.disa.mil/pki/eca/index.html>
A nominal fee is charged for each certificate. The Contractor, including any subcontractors, shall assume the responsibility for all costs of obtaining each digital certificate needed.
- After the required certificate is obtained:
 - (a) Click on the MHP-WARP hyperlink: <https://mhpwarp.redstone.army.mil/>
 - (b) Enter ECA password
 - (c) Click on Help and follow the instructions for obtaining the necessary access

ATTACHMENT 2 (continued)

- HELP Numbers are as follows:

MHP Access – (256)313-2143; DSN 897-2143

JMC Quality Administrators for WARP issues – (309)782-2697 or (309)782-7107

- Worldwide Ammunition-data Repository Program (WARP)

An online user's manual will provide additional help and criteria in the development of an ammunition data card. It is recommended that you download and read the user's manual prior to creating and filling out your initial ammo data card. The user's manual also contains screen shots, which depict what the data entry person will see during the ADC input process.

- Ammunition Data Card Input

ADC input allows current contractors and government facilities the capability to create, and submit for approval, both sample ADCs and production lot ADCs which meet the format requirements of MIL-STD-1168 and as specified in this SOW for ADCs. ADCs are automatically forwarded to the respective Government Agency Responsible for Acceptance (GARA). The GARA in most cases is the Defense Contract Management Agency (DCMA) Quality Assurance Representative (QAR), who reviews contractor input for accuracy and completeness, and after updating the disposition code for the specific lot, submits the ADC to the database. The contractor's data entry person is granted access only to ADCs identified with its specific manufacturing code, as identified in MIL-HDBK-1461, Manufacturer's Symbols. The use of previously inputted ADCs through the TEMPLATE option significantly reduces input effort, while increasing accuracy and consistency of data.

- Email Notification

WARP provides immediate, automated notification to process participants when actions are required. When the producer has completed an ADC submission, an email message is routed to the GARA advising that an ADC awaits review and approval. If the GARA approves the ADC as submitted, the ADC is released to the WARP database and an email, with approved data card, is routed back to the originator. If the ADC requires modification or correction to conform with MIL-STD-1168 and/or the approved format and technical content as approved by the government, an email is provided to the ADC originator advising that corrective action is required prior to approval.

- Information Updates

It is important that the System Administrators are apprised when a producer receives a new contract. The producer shall notify usarmy.RIA.jmc.mbx.warp@mail.mil within 30 days after receipt of a new contract. Information to be included shall be the contract number, item, GARA, Manufacturer's identification symbol and the names of the individuals who will be inputting ADCs into the system. If you are a new producer and do not have a Manufacturer's identification symbol, you can obtain one by sending an email to usarmy.RIA.jmc.mbx.warp@mail.mil. The email must contain manufacturer's name, address where performance of the contract will take place, and a point of contact.

(End of statement of work)

CAD/PAD Item Marking Instruction

Prepared By
CAD/PAD Joint Program Office
NSWC IHD and AFLCMC/EBHJ

Version 1.6
Dated 06 March 2023



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References

NOTE: Use latest revision of all reference documents.

1. **MIL-STD-130** Department of Defense Standard Practice Identification Marking of U.S. Military Property
2. **MIL-STD-1168 (current revision)** Ammunition Lot Numbering and Ammunition Data Card
(Note section 3.1 for new lot assignment for “remanufactured CAD/PAD Components”)
3. **ISO/IEC 16022** Information technology - Automatic identification and data capture techniques - Data Matrix bar code symbology specification
4. **ISO/IEC 15434** Information technology - Automatic identification and data capture techniques - Syntax for high capacity ADC media
5. **ISO/IEC 15418** Information technology - Automatic identification and data capture techniques - GS1 Application Identifiers and ASC MH10 Data Identifiers and maintenance
6. **UNDER SECRETARY OF DEFENSE FOR ACQUISITION POLICY AND TECHNOLOGY** Department of Defense Guide to Uniquely Identifying Items version 1.6 (this document is available on line at <http://www.acq.osd.mil/dpap/pdi/uid/index.html> or from Defense Procurement & Acquisition Policy, 3060 Defense Pentagon, Room 3E1044, Washington, DC 20301-3060)
7. **DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT 252.211-7003** Item Identification and Valuation (Copies of this document are available on line at <http://farsite.hill.af.mil/> or from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954)
8. **AFSCM 21-556 Vol. 2** Julian Date format
9. **ANS MH10.8.2 - 2006** Data Application Identifier Standard
10. **ISO 28219** Labeling and Direct Product Marking with Linear Bar Code and Two-Dimensional Symbols

Figures

1. Determining IUID requirement
2. Sample UID Data Stream
3. Sample CAD/PAD Data Plate/Label
4. Sample of Etched Data
5. Sample CAD/PAD Small Item Data Plate/Label

1. Purpose.

The Under Secretary of Defense has implemented policy requiring DoD components meeting the requirements identified in Figure 1 to be marked with **Item Unique Identifiers** (IUID). Cartridge Actuated Devices and Propellant Actuated Devices (CAD/PAD) items, regardless of dollar value, are typically serially managed in aircraft maintenance data systems and therefore are applicable to the IUID requirement.

Additionally, the Air Force Aircrew Escape System Executive Working Group (AESEWG) and the egress system maintenance community has requested that CAD/PAD items be marked with Data Matrix ECC 200 on item data plates or items to enhance accuracy of aircraft maintenance data.

In accordance with MIL-STD-130, this marking instruction identifies marking requirements that meet the above objectives by tailoring Unique Identification (UID) requirements to include aircraft maintenance data requirements for both free text Human Readable Information (HRI) and Machine Readable Information (MRI), which are both mandated requirements for CAD/PAD items.

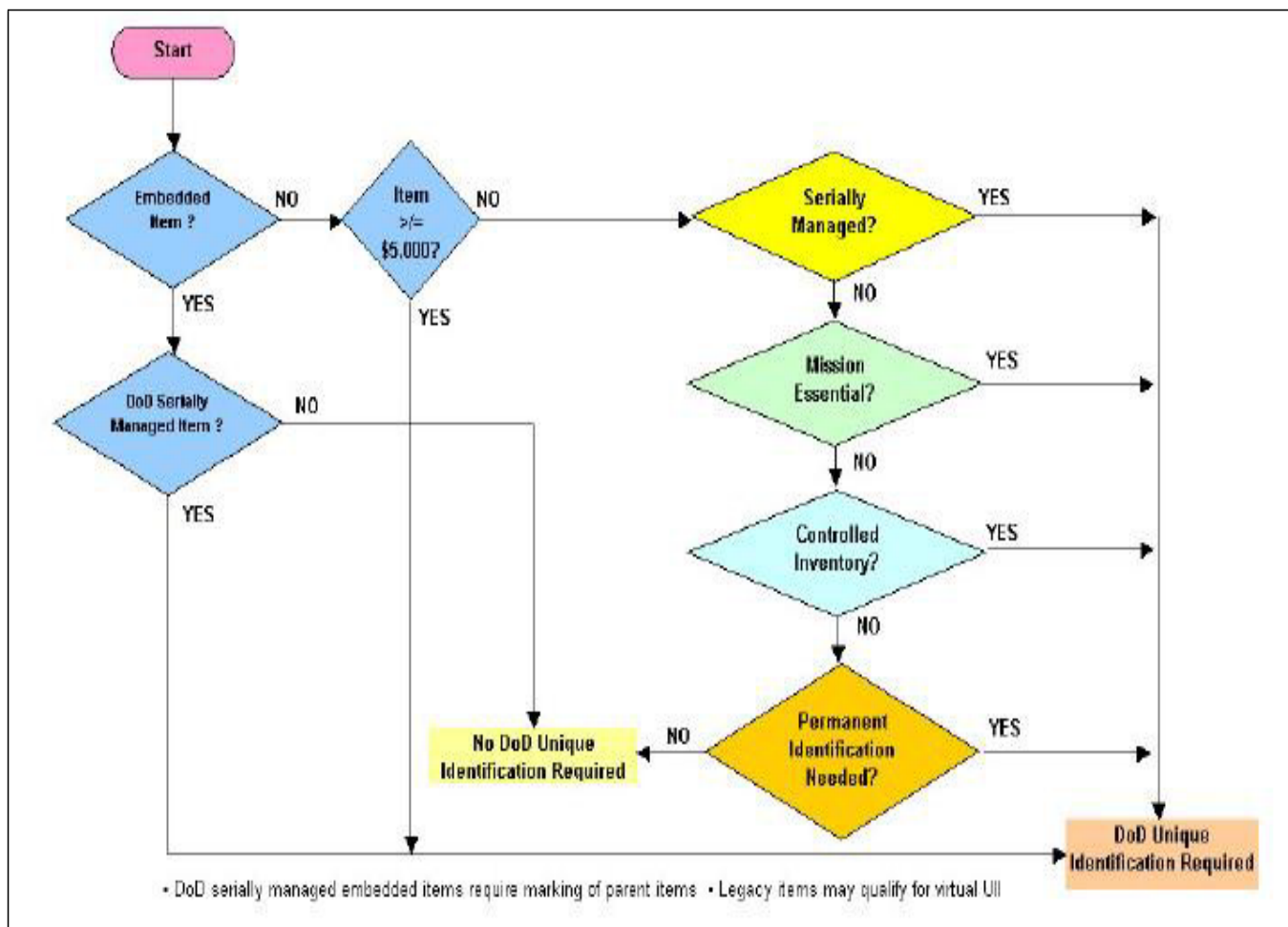


Figure 1. Determining UID Requirement

2. SCOPE.

This instruction identifies marking requirements applicable to suppliers/manufacturers of CAD/PAD components. It is applicable to both production of new CAD/PAD components and remanufacture of service returned, over-aged components. It identifies marking methods applicable to the various sizes and shapes of CAD/PAD components.

As of this writing, guidance regarding mandated transfer of data to the IUID Registry can be found at <https://dodprocurementtoolbox.com/site-pages/unique-id>.

3. **Marking Requirements.**

Both Human Readable Information (HRI) and Machine Readable Information (MRI) are required (see section 4). In accordance with the Defense Federal Acquisition Regulation Supplement (DFARS), DoD has mandated the use of Data Matrix ECC 200 as the standard format for UID MRI markings. Guidance for the construct and application of Data Matrix markings is contained in MIL-STD-130 and the DoD UID Guide located at <https://www.acq.osd.mil/asda/dpc/ce/ds/unique-id.html>. Training, guides and others resources are located on the left side under the IUID tab.

3. 1. **HRI Marking Requirements.** The following HRI data is required (as a minimum):

- Item Nomenclature; Per applicable manufacturer drawing
- Manufacturer's Commercial and Government Entity (CAGE); Five (5) alpha/numeric characters.
- Item Lot Number; Lot numbers shall be assigned IAW MIL-STD-1168 (current revision), unless otherwise directed. A new lot number and serial number shall also be assigned for "remanufactured" CAD/PAD components as opposed to adding a suffix to the original lot number for a "rework, overhaul or renovation" effort.

NOTE: The requirement for unique lot number has been deleted.

DOM is NOT required if the lot number contains the DOM as identified in MIL-STD-1168

- Item Serial Number; Serial number shall be up to eight (8) numeric characters. The serial numbers shall be sequentially assigned and shall not be repeated until serial number 99,999,999 has been reached.
- Item manufacturer's Part Number

NOTE: The weapon system Specification Number does not meet this requirement

- Optional markings: Additional markings (such as manufacturer's name, logo, contract number) are acceptable, provided that space is available on the item/data label. Date of Manufacture (DOM); DOM shall be MM/YYYY and be the same date that is coded into the lot number.

NOTE: Markings and/or attachments shall make no reference to item installation requirements/procedures or shelf/service lives

3.2. MRI Data Construct.

Data Matrix ECC200 markings shall be developed based on UID Data Construct #2 using Data Identifiers (DIs) in accordance with MIL-STD-130 and the DoD UID Guide (see Figure 2). To facilitate durability throughout the scanning life cycle, the largest symbol size, including quiet zone, shall be fitted within the available marking real estate. The symbol's internal module (cell) sizes shall be no smaller than 0.0075 inch (0.19 mm) and no larger than 0.025 inch (0.64 mm). Marks shall contain **only** the data elements identified below:

NOTE: The actual UII will be comprised of CAGE, Part Number, and Serial Number data. The parenthesis surrounding the Data Identifiers are NOT encoded in the 2D symbol

(17V) Manufacturer's Commercial and Government Entity (CAGE) code, 5 alpha/numeric characters

- (1P) Manufacturer's Item Part Number, 18 alpha/numeric characters maximum

NOTE: The weapon system specification numbers do not meet this requirement

- (S) Item Serial Number, IAW HRI requirement, up to eight (8) numeric characters

- (30T) Item production Lot Number, IAW HRI requirement, 15 alpha/numeric characters maximum

NOTE: The data identifier for Lot Number has been changed from 1T to 30T

- (4D) Date of Manufacture (DOM), in Julian Date format, five (5) numeric characters:

Example: 08151 = 31 May 2008

The first two positions are the last two digits of the year, the last three positions are the Julian Date (001 through 366)



DI> ^R_S 06 ^G_S 17V 12345 ^G_S 1P 123456789-ABCDEFGH ^G_S S 12345678 ^G_S 30T ABC06D123-001 ^G_S 4D 08151 ^R_S ^E_T ^O_T

Figure 2. Sample UID Data Stream

4. **Placement of UID Marking.** CAD/PAD items vary considerably in size and shape. A large percentage of these items are currently marked with HRI printed on data plates or labels permanently affixed to the item. The preferred method for incorporating the UID required MRI is to include it on the current data plate/label and, wherever possible, the supplier is to incorporate this concept. The following identifies known variations in marking and preferred MRI incorporation methods.

Note: Unless specifically identified as a requirement in the procurement contract, subcomponents of the End Item Assembly shall not be UID marked.

4.1. **CAD/PAD Items Currently Marked with Data Plates, Labels, or Flag-Tags.** The data plates/labels for these items shall be modified/arranged to allow space for the MRI per paragraph 3.2. Marking shall comply with MIL-STD-130 and the DoD UID Guide (see figure 3).



Figure 3. Sample Data Plate/Label

4.2. **CAD/PAD Items Currently Marked With Ink or Laser Etching Directly on the Item.** Item marking shall be modified/arranged to allow space for the incorporation of ink or laser etched MRI per paragraph 3.2. Marking shall comply with MIL-STD-130 and the DoD UID Guide (see figure 4).

CAD/PAD Builder Inc.
Acft Explosive Component Noun

CAGE: 12345
P/N: 123456789-1234
Ser No: 0001
Lot No: ABC08D001-001
DOM: 04/2008

Figure 4. Sample of Etched Data

4.3. **CAD/PAD Items Considered to be Too Small for HRI and MRI.** These items shall be physically marked using currently approved methods for HRI. A self-adhesive label shall be supplied in the item inner packaging that includes the minimum HRI (per paragraph 3.1) and the MRI Data Matrix marking (per paragraph 3.2). The label shall be as small as possible (meeting MIL-STD-130 requirements for HRI) and will include only the minimum required data (see figure 5).

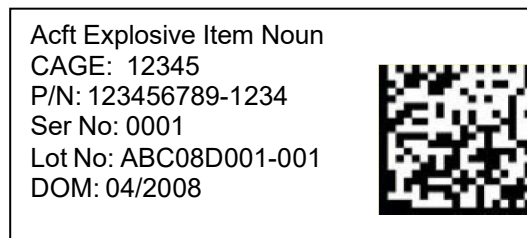


Figure 5. Sample Small Item Data Plate/Label

Note: This will allow the maintainer to scan the MRI from the label and the ability to affix it to the next higher assembly.

5. **Kit Marking Requirements.** Individual components procured and delivered in a “kit” or “ship-set” format shall each be marked with HRI and MRI as identified in sections 3 and 4. The kit containers shall also be marked with HRI and MRI per sections 3 and 4.

Note: Individual components supplied in kits or ship-sets require unique item identifiers. Their markings shall not reflect those identifying the kit or ship-set.

5.1 Only the “kit” or “ship-set” will require data submission into the UID Registry. Data submission for individual components in a kit is not required, as these components were not individually priced as line items when placed on contract.

Note: To meet UID requirements, each kit will be assigned an individual serial number. Kits that are comprised of multiple containers will all be marked with the same serial number.

5.2 “Kits” or “ship sets” will typically be assigned lot numbers coinciding with the oldest individual component in the kit.

6. **JSF F-35 Instructions.** The marking instructions for F-35 components, labels, and packaging, shall be as outlined in this document. Additionally, contractors are required to submit an Electronic Equipment Logbook (EEL) for all serial numbers delivered to the Government. When submitting EELs, the lot number is limited to 10 characters and shall only include the year of production, month of production, interfix number, and sequence number with no dashes. Use of these nine characters will allow for an optional lot suffix as

deemed necessary by procuring agency. The lot number identified on the Ammo Data Card (ADC) shall match the lot number used on the components and shall be generated IAW MIL-STD-1168.

7. **Waivers**. Waivers to the UID requirement are not allowed. Requests for waivers to marking methods outlined in this instruction may be considered. Submit requests through appropriate contracting channels.

RFV Form DD1694 Example Only dated 06/05/2025

REQUEST FOR VARIANCE (RFV)		1. DATE PREPARED: (DD-MMM-YYYY) 14-FEB-2024		2. RFV NUMBER: 24CK999-V001		OMB No. 0704-0188 OMB approval expires October 31, 2019	
				3. PROCURING ACTIVITY NO. (PAN): N00104			
The public reporting burden for this collection of information, 0704-0188, is estimated to average 2 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding the burden estimate or burden reduction suggestions to the Department of Defense, Washington Headquarters Services, at whs.mc-alex.esd.mbx.dd-dod-information-collections@mail.mil. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number. PLEASE DO NOT RETURN YOUR COMPLETED FORM TO THIS ADDRESS. RETURN COMPLETED FORM TO THE ACQUIRER ISSUING CONTRACTING OFFICER FOR THE CONTRACT/PROCURING ACTIVITY NUMBER LISTED IN BLOCK 22 OF THIS FORM.							
DISTRIBUTION STATEMENT: Verify distribution statement with the contract. DISTRIBUTION STATEMENT D: Distribution authorized to Department of Defense and U.S. DoD contractors only; Critical Technical Information (CTI); January 1, 2024. Other requests for this document shall be referred to Commanding Officer or Technical Director, NSWC Indian Head Division. Verify the date the distribution reason was assigned with the contract. If not in the contract, the date should match the date of the document.							
4. TITLE OF VARIANCE: JM60 Catapult battery voltage check Brief description of intent/purpose to include identifying info such as DODIC/model/type							
5. VARIANCE PRE OR POST-PRODUCTION:				6. BASELINE AFFECTED:			
☒ Pre-Production ☐ Post-Production				☐ Functional ☐ Allocated ☒ Product			
7. SYSTEM INFORMATION		7.a. MODEL/TYPE DESIGNATION: CKU-5C/A (JM60)		7.b. SYSTEM/CONFIG. ITEM NOMENCLATURE: F-16, F-22		7.c. END ITEM CAGE CODE: 14083 End item cage code	
8. AFFECTED ITEM NOMENCLATURE: Inflation Device, Water Activated FLU-11/A		9. PART NUMBER(S) OF AFFECTED ITEM(S): 855AS110, 1812-172-01 Per Section B of the contract ("See attached" if cont. page if needed). Nomenclature name should match affected item name.					
10.a. OTHER EXTERNAL SYSTEM AFFECTED?: ☒ Yes ☐ No		10.b. IF YES, LIST OTHER SYSTEMS OR CONFIGURATION ITEMS AFFECTED: Training System XXX If needed, enter the appropriate title or reference number of the supplier document that details/lists the defect numbers, or identifies the variance classification based on the defect number definition(s), when appropriate. Otherwise, disregard and enter N/A.					
11. IDENTIFICATION OF VARIANCE		11.a. VARIANCE CLASSIFICATION: ☐ Critical ☒ Major ☐ Minor		11.b. DEFECT NO. (if applicable): N/A		11.c. DOCUMENT DEFINING DEF (if applicable): N/A If needed, Enter the supplier defect number, which corresponds with the variance listing of defect characteristic(s) from which an authorized variance can be released, when appropriate. Otherwise, disregard and enter N/A.	
12. DESCRIPTION OF VARIANCE: Add battery voltage check after NO-FIRE SENSITIVITY. See attachment. Describe the nature of the proposed include marked drawings and/or other relevant documents.							
13. NEED FOR VARIANCE: Voltage needs to be checked after electrical test and before functional tests to ensure the No-Fire tests did not reduce the battery voltage below required volts.							
14. CORRECTIVE ACTION TAKEN: ECPXX-XXX is in the process of being approved to update the TDP. Describe the corrective actions taken or will be taken to prevent a future recurrence of this variance.							
15. EFFECT ON PERFORMANCE, FUNCTION, RELIABILITY, DURABILITY, INTEGRATED LOGISTICS SUPPORT, INTERFACE, OR SOFTWARE: None. Block 16a is only marked if the same request has been approved in the past. Block 16b will include the approved RFV number of the previous request. Approved RFVs should rarely be submitted on a recurring basis.							
16. RECURRENCE		16.a. RECURRING VARIANCE?: ☒ Yes ☐ No		16.b. IF YES, PROVIDE RATIONALE: Prior RFV XXXX-XXX was limited to Lot K099-12222. ECP was drafted at the same time, but not yet approved and implemented.			
17. EFFECTIVITY: (Quantity Affected, Lot Numbers Affected, Serial Numbers, Dates) Lots K999-12345 and K999-67890 Add affected, quantity, Lot Numbers and serial numbers Block 18 & 19 Enter the per unit cost/ impact of the variance, and if accepted post-production, a reduction in cost (enclosed in parentheses) should be presented. If needed							
18. PER UNIT COST IMPACT: \$0.00		19. TOTAL COST IMPACT: \$0.00		20. EFFECT ON SCHEDULE IF APPROVED: No impact to the schedule Describe any changes to the schedule dependent on the final disposition for the variance. If the per-unit cost in Box 18 and the total cost in Box 19 are both \$0.00, then Box 20 should indicate that there will be no cost impact.			
21. CONTRACT INFORMATION		21.a. SUPPLIER: ABC Manufacturing, Inc.		21.b. CONTRACT NO. AND LINE ITEM: N00104-23-C-K999 CLIN 001			
22. CONTRACTING OFFICER		22.a. NAME: Jane Doe		22.b. TELEPHONE: +1 (555) 555-5555		22.c. E-MAIL: jane.doe@navy.mil	

